

Design and Implementation of Vehicle-to-Vehicle (V2V) Communication System for Collision Avoidance

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ABSTRACT

In modern automotive systems, Vehicle-to-Vehicle (V2V) communication has emerged as a critical technology for enhancing road safety and preventing collisions. This project presents the design and implementation of a scaled-down educational V2V communication system that demonstrates the core concepts of real-world automotive V2V technology while operating within the constraints of a student prototype.

The system implements a three-board architecture using STM32F407VGT6 Discovery boards: Board 1 acts as a transmitter simulating a leading vehicle, Board 2 serves as a receiver/bridge combining wireless V2V data with ultrasonic sensor readings, and Board 3 functions as the main controller managing motor control, LED indicators, and buzzer alerts for collision warnings.

The project initially attempted to use NRF24L01+ modules for 2.4 GHz wireless communication but encountered hardware issues with the modules. Subsequently, the implementation was migrated to ESP32 modules using the ESP-NOW protocol, which provided more reliable communication between Board 1 and Board 2. The system also integrates CAN bus communication using MCP2551 transceivers for inter-board data transfer.

This project demonstrates the fundamental principles of V2V communication systems as specified in industry standards like DSRC (Dedicated Short-Range Communications), C-V2X (Cellular Vehicle-to-Everything), IEEE 802.11p, and SAE J2735. While operating at significantly lower data rates and ranges compared to industrial implementations, this prototype serves as an educational platform for understanding the core concepts of automotive V2V communication, particularly relevant given India's 2026 V2V mandate for all new vehicles.

Keywords: V2V Communication, STM32F407VGT6, ESP32, ESP-NOW, CAN Bus, Collision Avoidance, DSRC, C-V2X, Automotive Embedded Systems

1. INTRODUCTION

1.1 Introduction to V2V Communication

Vehicle-to-Vehicle (V2V) communication is a transformative technology that enables automobiles to exchange critical safety and operational data wirelessly. This technology forms a crucial component of the broader Intelligent Transportation Systems (ITS) ecosystem, enabling vehicles to share information about their speed, position, heading, and other parameters in real-time.

The fundamental principle behind V2V communication is to create an awareness bubble around each vehicle, allowing it to perceive potential hazards beyond the line of sight. This capability is particularly valuable in scenarios such as intersection collision warning, emergency vehicle approach notification, forward collision warning, and blind spot detection.

In the context of Indian automotive regulations, the Government of India has mandated that all new vehicles sold after 2026 must be equipped with V2V communication capabilities. This regulatory push makes V2V technology a highly relevant area for embedded systems engineers entering the automotive industry.

This project aims to create a functional prototype that demonstrates the core concepts of V2V communication, including wireless data transmission, collision detection algorithms, and warning systems. While the prototype operates at a smaller scale compared to industrial implementations, it provides valuable insights into the architecture and challenges of real-world V2V systems.

1.2 Literature Survey

The literature on V2V communication encompasses diverse perspectives, addressing communication protocols, safety applications, and regulatory frameworks.

DSRC (Dedicated Short-Range Communications): DSRC is the established standard for V2V communication, operating in the 5.9 GHz band (5.850-5.925 GHz). Based on IEEE 802.11p (now IEEE 802.11bd), DSRC provides data rates of 3-27 Mbps with a typical range of 300-1000 meters and latency under 100 milliseconds.

C-V2X (Cellular Vehicle-to-Everything): C-V2X is an emerging alternative that leverages cellular network infrastructure. Operating in the 5.9 GHz band for direct communication (PC5 mode) and cellular bands for network-assisted communication (Uu mode), C-V2X offers improved range and reliability compared to DSRC.

SAE J2735 Basic Safety Message: The SAE J2735 standard defines the message format for V2V communication, including the Basic Safety Message (BSM) that contains vehicle position, speed, heading, acceleration, and other critical data. BSMs are broadcast at a rate of 10 Hz (10 messages per second).

Research on Collision Avoidance: Studies have explored various algorithms for Time-to-Collision (TTC) calculation, including constant velocity models, constant acceleration models, and more sophisticated trajectory prediction methods. These algorithms form the basis for collision warning systems.

Sensor Fusion Approaches: Research indicates that combining V2V communication data with onboard sensors (radar, lidar, cameras) provides more reliable collision detection than either technology alone. This sensor fusion approach is becoming standard in advanced driver assistance systems (ADAS).

1.3 Aim of Project

The project aims to design and implement a scaled-down V2V communication system that demonstrates the fundamental principles of automotive V2V technology. The system uses STM32F407VGT6 Discovery boards as the primary microcontroller platform, with wireless communication initially attempted using NRF24L01+ modules and subsequently implemented using ESP32 modules with ESP-NOW protocol.

The primary objectives of this project are:

- To understand and implement the core concepts of V2V communication
- To design a three-board architecture that simulates vehicle-to-vehicle data exchange
- To implement wireless communication using available 2.4 GHz modules
- To integrate CAN bus communication for inter-board data transfer
- To implement collision detection algorithms with visual and audio warnings
- To compare the prototype implementation with industry standards

1.4 Scope and Objective

Scope:

- Design and implementation of wireless V2V communication using ESP32 ESP-NOW protocol
- Integration of ultrasonic sensors (HC-SR04) for distance measurement
- Implementation of CAN bus communication using MCP2551 transceivers
- Motor control using L293D drivers for vehicle simulation
- LED and buzzer-based warning system for collision alerts
- Vehicle simulation including speed, lane changes, and emergency braking scenarios

Objectives:

- Demonstrate real-time data exchange between simulated vehicles
- Implement Time-to-Collision (TTC) calculations for collision warning
- Create a functional prototype suitable for educational demonstration
- Document the implementation for team sharing and interview preparation
- Understand the gap between student prototypes and industrial V2V systems

2. THEORETICAL DESCRIPTION OF PROJECT

2.1 V2V Communication Standards (Industry)

Understanding industry standards is crucial for appreciating the design decisions made in this prototype. The following are the key standards governing V2V communication:

IEEE 802.11p (WAVE - Wireless Access in Vehicular Environments): This standard defines the physical and MAC layer specifications for vehicular communication. It operates in the 5.9 GHz band with a channel bandwidth of 10 MHz, providing data rates from 3 to 27 Mbps.

IEEE 1609 Family: This suite of standards defines the network and application layers for DSRC/WAVE systems, including security services, networking services, and resource management.

SAE J2735: This standard defines the message sets used in V2V communication, including the Basic Safety Message (BSM) which contains vehicle position (latitude, longitude, elevation), motion data (speed, heading, acceleration), and vehicle status information.

3GPP C-V2X (Release 14 and beyond): The cellular V2X standard provides an alternative to DSRC, leveraging LTE and 5G technologies for vehicle communication. It supports both direct communication (PC5 interface) and network-based communication (Uu interface).

2.2 Communication Technologies

NRF24L01+ (Initially Attempted): The NRF24L01+ is a 2.4 GHz transceiver module operating in the ISM band (2.4-2.525 GHz). It supports data rates of 250 Kbps, 1 Mbps, or 2 Mbps with a theoretical range of up to 100 meters in open space. The module

communicates with the microcontroller via SPI interface and requires 3.3V power supply with proper capacitor placement for stable operation.

ESP32 with ESP-NOW Protocol (Final Implementation): ESP-NOW is a proprietary wireless protocol developed by Espressif Systems. It enables direct device-to-device communication without the need for WiFi infrastructure. ESP-NOW supports up to 250 bytes per message with low latency (typically < 10ms) and operates in the 2.4 GHz band. It is particularly suitable for short-range, low-power applications.

CAN Bus (Controller Area Network): CAN is a robust serial communication protocol originally designed for automotive applications. The MCP2551 transceiver is used to interface the STM32 CAN controller with the physical bus. CAN supports data rates up to 1 Mbps and provides built-in error detection and handling mechanisms.

2.3 Hardware Components

STM32F407VGT6 Discovery Board: This is a 32-bit ARM Cortex-M4 microcontroller operating at up to 168 MHz. It features 1 MB Flash memory, 192 KB SRAM, and a rich set of peripherals including SPI, UART, CAN, and GPIO. The Discovery board includes an on-board ST-LINK/V2 debugger for programming and debugging.

ESP32 Module: The ESP32 is a dual-core 240 MHz microcontroller with integrated WiFi and Bluetooth. For this project, it is used primarily for its ESP-NOW wireless communication capability. The module operates at 3.3V and interfaces with the STM32 via UART.

HC-SR04 Ultrasonic Sensor: This sensor provides distance measurement using ultrasonic pulses. It has a measuring range of 2 cm to 400 cm with an accuracy of approximately 3 mm. The sensor requires a 10 μ s trigger pulse and provides an echo pulse proportional to the measured distance.

MCP2551 CAN Transceiver: This is a high-speed CAN transceiver that provides the physical interface between the STM32 CAN controller and the CAN bus. It supports data rates up to 1 Mbps and includes features for bus fault protection.

L293D Motor Driver: This dual H-bridge motor driver IC is used to control DC motors for vehicle simulation. It can drive two DC motors with currents up to 600 mA per channel.

2.4 Protocol Overview

ESP-NOW Protocol Structure: ESP-NOW uses a vendor-specific action frame format. Each frame contains a header with device MAC addresses, followed by the payload data (up to 250 bytes). The protocol supports both unicast and broadcast communication modes.

V2V Data Message Structure (This Project): Our V2V message contains: Vehicle ID (4 bytes), Speed (2 bytes), Relative Distance (2 bytes), Lane Position (1 byte), Brake Status (1 byte), Turn Signal (1 byte), Emergency Status (1 byte), and Timestamp (4 bytes).

CAN Message Format: CAN frames consist of an identifier (11-bit standard or 29-bit extended), control field, data field (0-8 bytes), and CRC for error checking. This project uses standard 11-bit identifiers with 8-byte data payloads.

3. SYSTEM DESIGN AND IMPLEMENTATION

3.1 System Architecture

The system employs a three-board architecture where each board serves a specific function in the V2V communication chain:

[SYSTEM ARCHITECTURE BLOCK DIAGRAM TO BE INSERTED HERE]

Board 1 - Transmitter (Leading Vehicle Simulation):

- Simulates a leading vehicle generating realistic driving data
- Generates vehicle speed, lane position, brake status, and emergency conditions
- Transmits V2V data via ESP32 ESP-NOW protocol
- Speed range: 0-120 km/h (simulated), with lane change and emergency braking scenarios

Board 2 - Receiver/Bridge (Following Vehicle Interface):

- Receives V2V data from Board 1 via ESP32 ESP-NOW
- Integrates ultrasonic sensor (HC-SR04) data for distance measurement
- Combines wireless V2V data with local sensor data
- Forwards combined data to Board 3 via CAN bus

Board 3 - Main Controller (Decision and Actuation):

- Receives combined data from Board 2 via CAN bus
- Performs collision detection algorithms and TTC calculations
- Controls LED indicators for warning levels (Green/Yellow/Red)
- Activates buzzer for audio warnings
- Controls motors via L293D for vehicle simulation

3.2 Three-Board Architecture Diagram

[THREE-BOARD ARCHITECTURE DIAGRAM TO BE INSERTED HERE]

The data flow in the system follows this sequence:

1. Board 1 generates simulated vehicle data (speed, position, brake status)
2. ESP32 on Board 1 transmits data via ESP-NOW protocol
3. ESP32 on Board 2 receives data and forwards to STM32 via UART

4. Board 2 STM32 reads ultrasonic sensor for local distance measurement
5. Board 2 combines V2V data with sensor data and transmits via CAN bus
6. Board 3 receives CAN data and performs collision detection
7. Board 3 activates appropriate warnings and controls motors

3.3 Software Design

Development Environment: STM32CubeIDE was used for STM32 firmware development, with Arduino IDE for ESP32 programming. PuTTY was used for serial debugging.

Board 1 State Machine: The transmitter operates in a simple loop: Initialize -> Generate Data -> Transmit -> Delay -> Repeat. Data generation includes random variations within realistic ranges to simulate actual driving conditions.

[BOARD 1 FSM DIAGRAM TO BE INSERTED HERE]

Board 2 State Machine: States include: IDLE -> RECEIVE_WIRELESS -> READ_SENSOR -> COMBINE_DATA -> TRANSMIT_CAN -> IDLE. Error handling includes timeout detection and LED-based error indication.

[BOARD 2 FSM DIAGRAM TO BE INSERTED HERE]

Board 3 State Machine: States include: IDLE -> RECEIVE_CAN -> CALCULATE_TTC -> DETERMINE_WARNING_LEVEL -> ACTIVATE_OUTPUTS -> IDLE. Warning thresholds are configurable for different scenarios.

[BOARD 3 FSM DIAGRAM TO BE INSERTED HERE]

3.4 Signal Specifications

[SIGNAL SPECIFICATION TABLE TO BE INSERTED HERE - Similar to sample report]

4. DEBUGGING AND TROUBLESHOOTING

This section documents the significant debugging challenges encountered during the project and the systematic troubleshooting steps taken to address them. These experiences provide valuable insights into embedded systems development and hardware debugging.

4.1 NRF24L01+ Module Issues

Problem Description: The project initially planned to use NRF24L01+ modules for wireless communication between Board 1 and Board 2. However, the purchased modules exhibited a critical hardware defect - a short circuit between VCC and GND pins.

Symptoms Observed:

- Modules drew excessive current when powered
- No response on SPI interface
- Modules became hot during operation
- Continuity test with multimeter showed VCC-GND short

Debugging Steps Performed:

8. **Visual Inspection:** Examined modules under magnification for solder bridges or damaged components
9. **Continuity Testing:** Used multimeter to verify VCC-GND short circuit on all purchased modules
10. **Power Supply Verification:** Confirmed 3.3V supply was within specifications
11. **Capacitor Placement:** Added 10 μ F and 100nF capacitors at VCC pin as recommended
12. **SPI Signal Verification:** Used oscilloscope to verify SPI clock and data signals
13. **Multiple Module Testing:** Tested all available modules - all showed same defect

Conclusion: All NRF24L01+ modules purchased were found to be defective with manufacturing defects causing VCC-GND short circuits. Due to time constraints for ordering replacement modules, the decision was made to migrate to an alternative wireless solution.

4.2 Migration to ESP32 ESP-NOW

Decision Rationale: ESP32 modules were selected as an alternative due to their availability, built-in WiFi/ESP-NOW capability, and simpler integration requirements. ESP-NOW protocol was chosen for its low latency and point-to-point communication characteristics suitable for V2V applications.

Implementation Steps:

14. Obtained MAC addresses of both ESP32 modules using diagnostic sketch
15. Configured ESP-NOW peer registration with target MAC addresses
16. Implemented send callback for transmission confirmation
17. Implemented receive callback for incoming data handling
18. Defined V2V message structure matching the project requirements
19. Tested communication between two ESP32 modules - SUCCESSFUL

Result: ESP-NOW communication between Board 1 (ESP32 transmitter) and Board 2 (ESP32 receiver) was successfully established and tested. Data transmission is reliable with low latency.

[ESP32 TO ESP32 COMMUNICATION TEST SCREENSHOT TO BE INSERTED HERE]

4.3 Board 3 Communication Challenges

Problem Description: While ESP32-to-ESP32 communication works correctly, Board 3 (STM32F407) is not receiving data from Board 2. The communication chain between ESP32 on Board 2 and STM32 on Board 3 is not functioning as expected.

Symptoms Observed:

- Board 2 ESP32 receives data from Board 1 (confirmed via serial monitor)
- Board 3 STM32 does not show any indication of receiving CAN messages
- LED error indicators on Board 3 remain in initial state
- No CAN interrupt triggers on Board 3

4.4 Debugging Steps Performed for Board 3

Step 1: UART Communication Verification (ESP32 to STM32 on Board 2)

- Added debug print statements on ESP32 to confirm UART transmission
- Verified UART baud rate matching (115200 baud)
- Checked TX/RX pin connections and ensured proper level shifting (3.3V)
- Used oscilloscope to verify UART signal waveforms

Step 2: STM32 UART Reception on Board 2

- Verified STM32CubeMX UART configuration (baud rate, word length, parity)
- Enabled UART interrupt and verified NVIC configuration
- Added LED toggle in UART receive interrupt to confirm reception
- Implemented ring buffer for UART data handling

Step 3: CAN Bus Configuration on Board 2 (Transmitter)

- Verified CAN peripheral initialization in STM32CubeMX
- Checked CAN bit timing parameters (prescaler, BS1, BS2)
- Verified CAN_TX and CAN_RX pin alternate function configuration
- Added HAL_CAN_GetTxMailboxesFreeLevel() check before transmission
- Verified transmission using HAL_CAN_AddTxMessage() return value

Step 4: CAN Bus Hardware Verification

- Checked MCP2551 transceiver power supply (5V)
- Verified CAN_H and CAN_L connections between boards
- Checked 120Ω termination resistors at both ends of CAN bus
- Used oscilloscope to observe CAN bus waveforms during transmission
- Measured differential voltage between CAN_H and CAN_L

Step 5: CAN Bus Configuration on Board 3 (Receiver)

- Verified CAN peripheral clock is enabled
- Checked CAN bit timing matches Board 2 configuration
- Verified CAN filter configuration (accept all messages for testing)
- Enabled CAN RX FIFO0 message pending interrupt
- Verified HAL_CAN_Start() is called after initialization
- Verified HAL_CAN_ActivateNotification() for RX interrupt

Step 6: Loopback Testing

- Configured Board 3 CAN in loopback mode
- Transmitted test message and verified reception in same board
- Loopback test helps isolate whether issue is in STM32 configuration or external bus

Step 7: Error State Analysis

- Checked CAN Error Status Register (ESR) for error flags
- Monitored Transmit Error Counter (TEC) and Receive Error Counter (REC)
- Checked for Bus-Off state or Error Passive state
- Added error callback handlers to capture CAN errors

Step 8: LED-Based Diagnostic Pattern

- Implemented LED patterns to indicate different states:
 - All LEDs OFF: System not initialized
 - Green blinking: System initialized, waiting for data
 - Yellow blinking: Data being processed
 - Red blinking: Error detected
 - Pattern sequence: Specific error code indication

Current Status: Despite extensive debugging, the communication between Board 2 and Board 3 remains unresolved at the time of report submission. The ESP32-to-ESP32 V2V communication (Board 1 to Board 2) works successfully, demonstrating the core V2V communication concept.

5. INDUSTRY VS PROTOTYPE COMPARISON

This section provides a detailed comparison between industry-grade V2V systems and our student prototype, highlighting both the limitations of our implementation and demonstrating our understanding of real-world requirements.

5.1 Industry Standards (DSRC, C-V2X)

DSRC (Dedicated Short-Range Communications):

- Frequency Band: 5.850-5.925 GHz (75 MHz bandwidth)
- Standard: IEEE 802.11p (WAVE) / IEEE 802.11bd
- Data Rate: 3-27 Mbps
- Range: 300-1000 meters
- Latency: < 100 ms (typically 20-50 ms)
- Message Rate: 10 Hz (10 BSMs per second)

C-V2X (Cellular Vehicle-to-Everything):

- Frequency Band: 5.9 GHz (PC5) + cellular bands (Uu)
- Standard: 3GPP Release 14/15/16
- Data Rate: Up to 1 Gbps (with 5G NR)
- Range: > 1000 meters (direct mode)
- Latency: < 20 ms (with 5G NR)

5.2 Data Rate Comparison

Parameter	Industry (DSRC/C-V2X)	Our Prototype (ESP-NOW)
Data Rate	3-27 Mbps (DSRC)	~1 Mbps
Frequency Band	5.9 GHz	2.4 GHz
Range	300-1000 meters	50-200 meters
Latency	< 100 ms	< 10 ms (typical)
Message Rate	10 Hz (100 ms interval)	5-10 Hz (100-200 ms)
Message Size	~300 bytes (BSM)	16-20 bytes

Position Accuracy	< 1.5 m (with DGPS)	Simulated values
Security	PKI-based (IEEE 1609.2)	None (prototype)

5.3 Range and Latency Comparison

Industry V2V Speed-Distance Relationship:

At highway speeds (120 km/h = 33.3 m/s), vehicles cover approximately 3.3 meters per 100 ms message interval. With a range of 300+ meters, vehicles have approximately 9 seconds of warning time for potential collisions.

Our Prototype Speed-Distance Relationship:

Our prototype uses simulated speed values and operates within indoor/lab distances (typically < 10 meters). The demonstration shows the concept of V2V communication but does not replicate real-world driving scenarios.

Key Insight: While our prototype operates at a much smaller scale, it successfully demonstrates the fundamental concept of wireless V2V communication: exchanging vehicle state information between two nodes and using that information to calculate collision warnings.

6. RESULTS AND TOOLS USED

6.1 Working Components

Successfully Implemented:

- ESP32 to ESP32 wireless communication using ESP-NOW protocol
- V2V data message structure implementation
- Vehicle simulation data generation (speed, lane, brake status)
- STM32CubeIDE project configuration for all three boards
- CAN bus driver implementation
- LED-based diagnostic and warning system
- Ultrasonic sensor driver implementation

Partially Implemented (With Issues):

- CAN bus communication between Board 2 and Board 3 (debugging ongoing)
- End-to-end V2V communication chain (Board 1 -> Board 2 -> Board 3)

Not Implemented Due to Hardware Issues:

- NRF24L01+ wireless communication (modules defective)

6.2 Test Results

ESP32 to ESP32 Communication Test:

[SCREENSHOT: ESP32 TRANSMITTER SERIAL OUTPUT TO BE INSERTED HERE]

[SCREENSHOT: ESP32 RECEIVER SERIAL OUTPUT TO BE INSERTED HERE]

[SCREENSHOT: SUCCESSFUL V2V DATA EXCHANGE TO BE INSERTED HERE]

Test Results Summary:

- ESP-NOW link establishment: Successful
- Data packet transmission: Verified with send callback
- Data packet reception: Verified with receive callback
- Message integrity: All fields correctly received
- Latency: < 10 ms (measured with timestamp)

6.3 Tools Used

Software Tools:

- **STM32CubeIDE:** Integrated development environment for STM32 microcontrollers, including code editor, compiler, and debugger
- **STM32CubeMX:** Graphical tool for peripheral configuration and code generation
- **Arduino IDE:** Development environment for ESP32 programming
- **PuTTY:** Serial terminal for debugging and data monitoring
- **Fritzing/Draw.io:** Circuit and block diagram design

Hardware Tools:

- **Digital Multimeter:** Voltage, current, and continuity measurements
- **Oscilloscope:** Signal waveform analysis for UART, SPI, and CAN
- **Logic Analyzer:** Digital signal protocol analysis
- **Soldering Station:** Hardware assembly and modifications

7. CONCLUSION

This project successfully demonstrated the fundamental concepts of Vehicle-to-Vehicle (V2V) communication through the implementation of a scaled-down prototype using STM32F407VGT6 Discovery boards and ESP32 modules.

Key Achievements:

- Successfully implemented wireless V2V communication using ESP32 ESP-NOW protocol after overcoming hardware failures with NRF24L01+ modules
- Designed and implemented a three-board architecture that mirrors the conceptual structure of real V2V systems
- Developed comprehensive firmware for vehicle simulation, data transmission, and collision detection
- Gained practical experience in debugging embedded systems hardware and software
- Created detailed documentation suitable for team collaboration and knowledge transfer

Challenges Encountered and Lessons Learned:

- Hardware quality issues with NRF24L01+ modules highlighted the importance of component verification before integration
- CAN bus communication challenges demonstrated the complexity of multi-node embedded systems
- Systematic debugging approach using LED indicators and serial monitoring proved essential for troubleshooting
- The gap between student prototypes and industrial V2V systems provided valuable perspective on automotive industry requirements

Industry Relevance:

This project is particularly relevant given India's 2026 mandate for V2V communication in all new vehicles. While the prototype operates at a significantly smaller scale compared to industrial DSRC and C-V2X systems, it demonstrates understanding of the core concepts: wireless data exchange, collision detection algorithms, and warning systems. The experience gained in working with STM32 microcontrollers, wireless communication protocols, and CAN bus systems directly applies to automotive embedded systems development.

The project successfully bridges the gap between theoretical knowledge of V2V communication standards (IEEE 802.11p, SAE J2735, C-V2X) and practical implementation experience, making it valuable preparation for careers in the automotive embedded systems industry.

8. FUTURE SCOPE

The V2V communication system developed in this project can be expanded in several directions:

Enhanced Wireless Communication:

- Migration to 5.9 GHz DSRC modules for compliance with automotive standards
- Implementation of C-V2X using 4G/5G cellular modules
- Increased range and reliability for outdoor testing

GPS Integration:

- Integration of GPS modules for actual position tracking
- Implementation of differential GPS (DGPS) for sub-meter accuracy
- Real-world testing with actual vehicle positions

Security Implementation:

- Implementation of message authentication using digital signatures
- Certificate-based identity management (IEEE 1609.2 compliance)
- Privacy protection through pseudonym certificates

Multi-Vehicle Network:

- Scaling to support multiple vehicles in a mesh network
- Implementation of VANET (Vehicular Ad-hoc Network) protocols
- Multi-hop communication for extended range

Advanced Collision Avoidance:

- Integration of radar and camera sensors for sensor fusion
- Machine learning-based trajectory prediction
- Autonomous emergency braking integration

V2X Extension:

- Vehicle-to-Infrastructure (V2I) communication with traffic signals
- Vehicle-to-Pedestrian (V2P) communication for vulnerable road user protection
- Vehicle-to-Network (V2N) for cloud connectivity

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